Symbolic Regression for Energy Efficiency Prediction: A Comparative Analysis with Traditional Machine Learning Models

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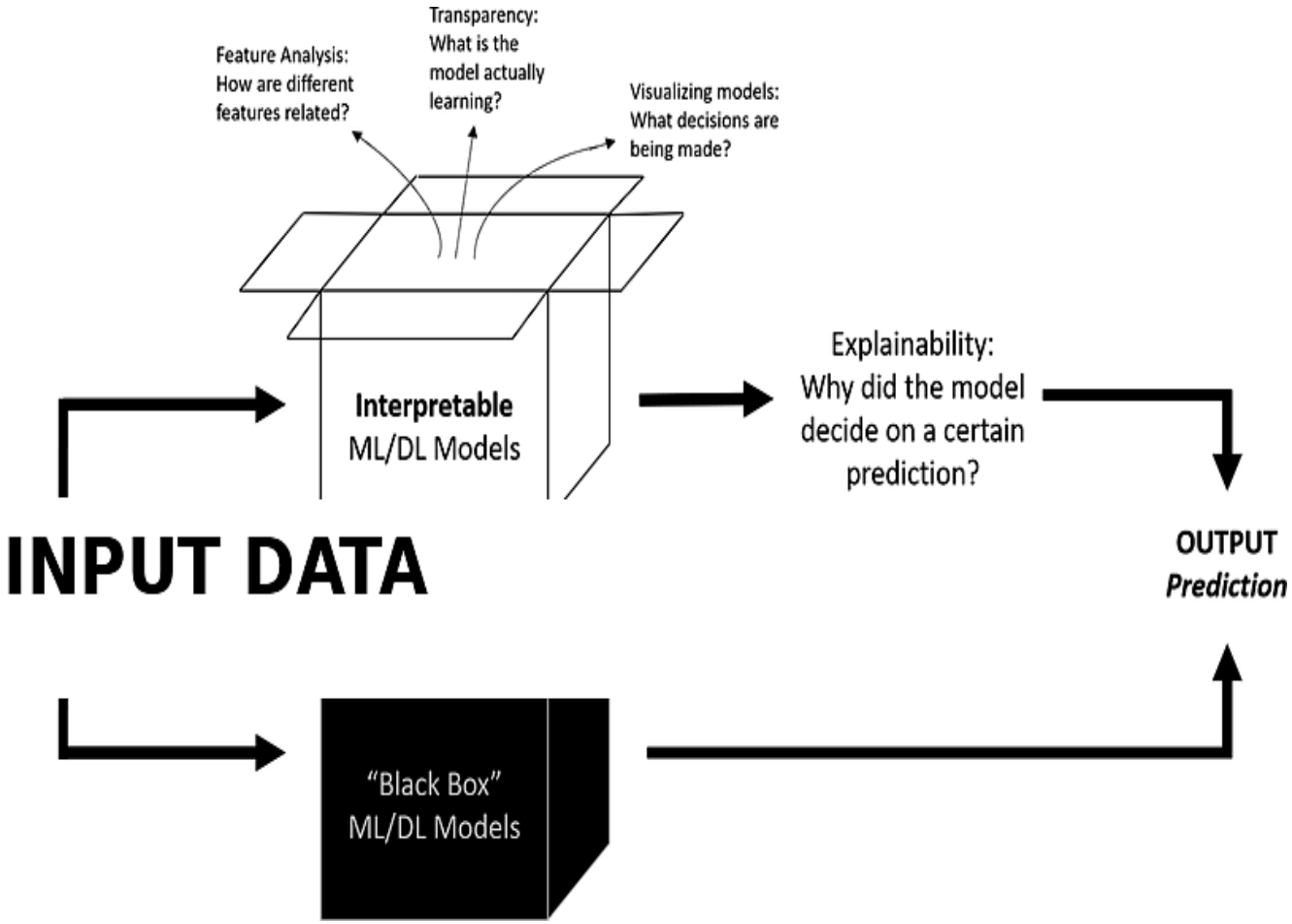
Abstract

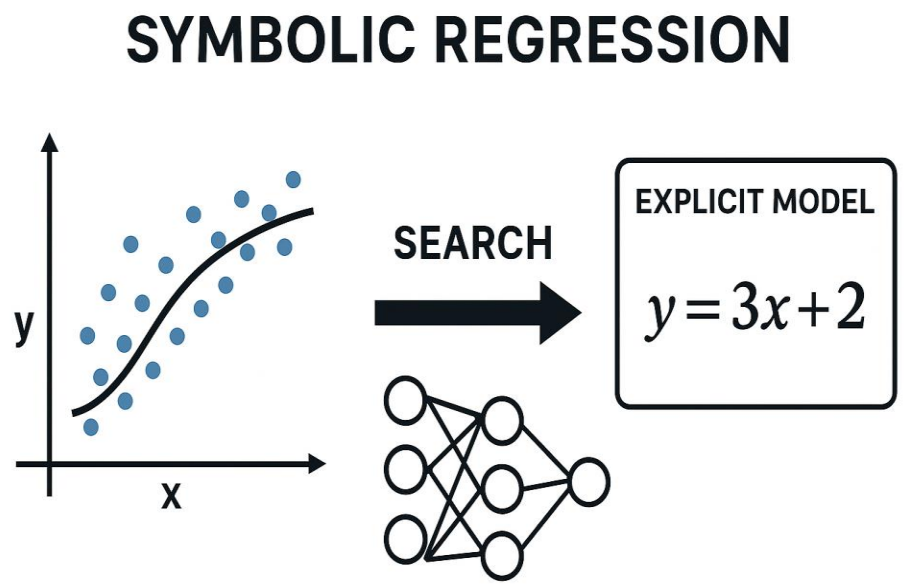
This project presents a hybrid framework for predicting **building heating load** using **traditional machine learning** and **symbolic regression** techniques. While conventional regression models provide high predictive accuracy, they lack **interpretability**. Symbolic regression is employed to discover an explicit **mathematical equation** representing the relationship between building design parameters and heating load. The discovered equation is integrated into a real-time interactive simulation platform for visualization and optimization. The system enables interpretable, deployable, and optimization-ready energy modeling for engineering applications.

Introduction

Symbolic Regression is an advanced **machine learning** technique that discovers **mathematical equations** directly from data without assuming a predefined model structure. Unlike **traditional regression** methods such as **Linear or Ridge Regression**, it searches for the best-fitting equation by exploring combinations of **mathematical operators and variables**. The goal is to generate an **interpretable equation** that explains the relationship between **input variables** and the **target output**. Unlike traditional regression methods, Symbolic Regression provides **transparent, human-readable models,** making it especially useful for understanding complex patterns and uncovering hidden relationships in data.

This project introduces a **hybrid symbolic regression** framework for interpretable **heating load modeling**. Symbolic regression is employed to automatically discover explicit mathematical equations that describe the relationship between **building design features** and **heating load**. The discovered equation is integrated into a real-time interactive simulation platform, enabling visualization, sensitivity analysis, and optimization of building parameters. By combining predictive modeling, interpretability, and deployment, the proposed system transforms energy prediction into an explainable and decision-support engineering tool.


Feature Analysis:
How are different
features related?
Transparency:
What is the
model actually
learning?
Visualizing models:
What decisions are
being made?
Interpretable
ML/DL Models
Explainability:
Why did the model
decide on a certain
prediction?
INPUT DATA
OUTPUT
Prediction
"Black Box"
ML/DL Models


SYMBOLIC REGRESSION
y
x
SEARCH
EXPLICIT MODEL
$y = 3x + 2$

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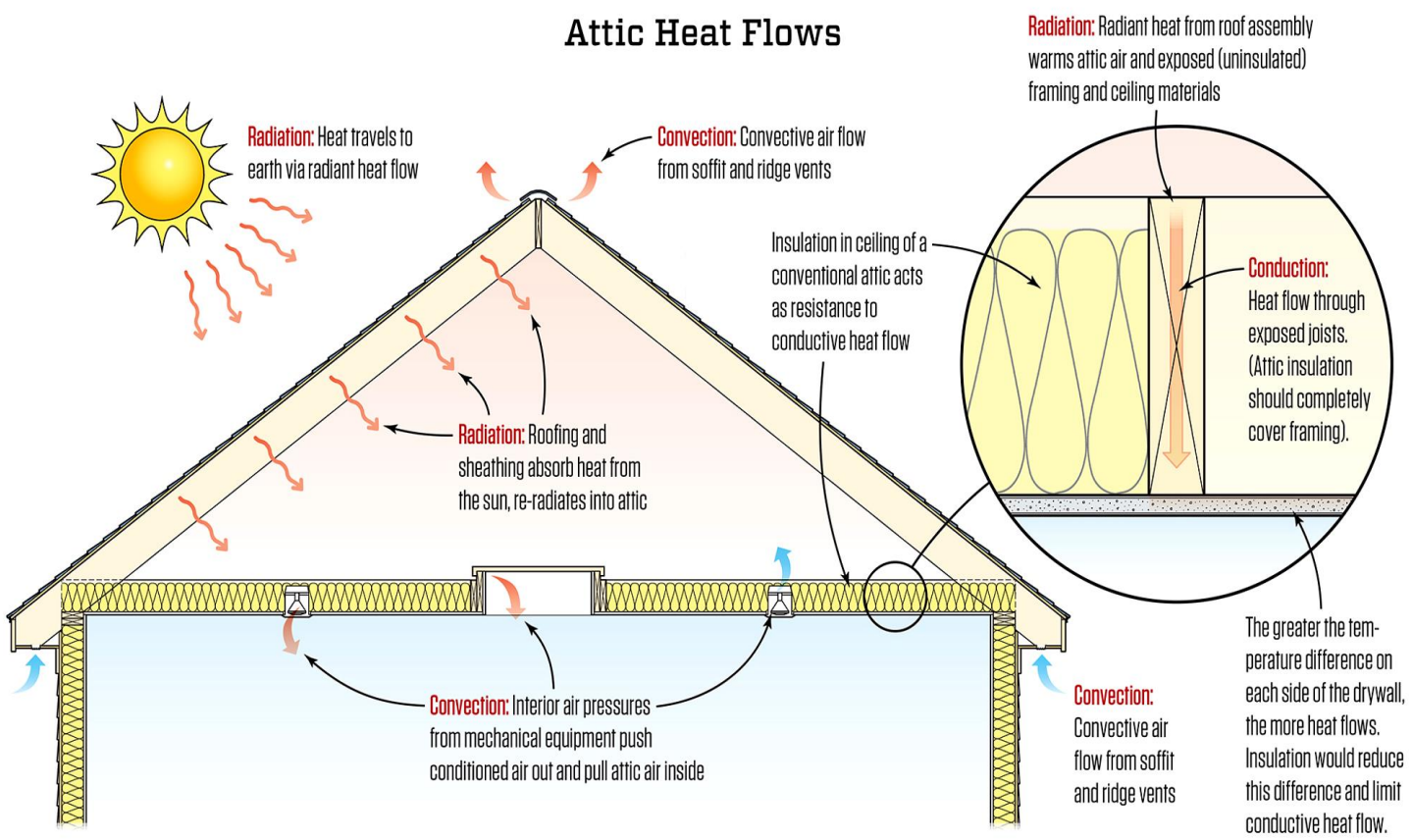


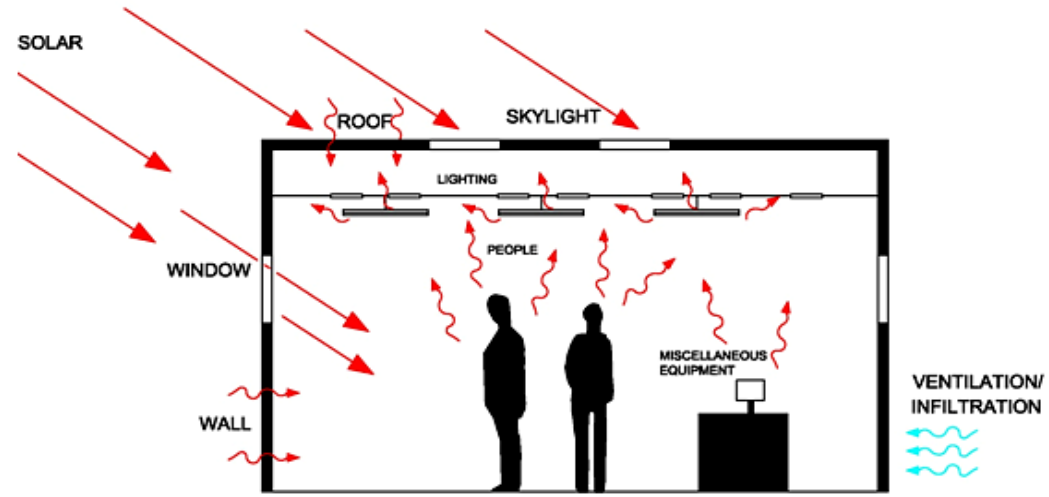


FIGURE 5: VARIOUS HEAT GAINS IN A BUILDING

External	Internal
Roofs/Walls – Conduction	Lights
Roofs/Walls – Radiation	People
Skylights/Windows – Conduction	Miscellaneous Equipment
Skylights/Windows – Radiation	
Ventilation/Infiltration	

Literature Survey

S.No	Paper Title / Year	Authors	Core Idea / Contribution	Relevance to Current Work
1	Accurate Quantitative Estimation of Energy Performance of Residential Buildings Using Statistical Machine Learning Tools (2012)	Tsanas & Xifara	Applied machine learning models to predict heating and cooling load using building design parameters.	Directly uses the same Energy Efficiency dataset used in our project.
2	Discovering Symbolic Models from Deep Learning with Inductive Biases (2020)	Cranmer et al.	Combines deep learning and symbolic regression to discover interpretable mathematical models from data.	Inspired interpretable equation discovery in our symbolic regression framework.
3	Interpretable Scientific Discovery with Symbolic Regression: A Review (2024)	Makke & Chawla	Comprehensive survey of modern symbolic regression techniques and applications in scientific modeling.	Provided theoretical foundation for symbolic regression methodology.

4	Symbolic Regression Boosting (2021)	Sipper & Moore	Introduced boosting techniques to improve symbolic regression predictive performance.	Demonstrates improvements in SR model accuracy and robustness.
5	Generalizability Improvement of Interpretable Symbolic Regression Models (2024)	Shirasawa et al.	Proposed methods to enhance generalization capability of symbolic regression models.	Inspired evaluation of generalization performance in our model.
6	A Flexible Symbolic Regression Method for Constructing Interpretable Clinical Prediction Models (2023)	La Cava et al.	Introduced flexible symbolic regression models for interpretable predictive analytics.	Motivated use of interpretable equations instead of black-box models.
7	Symbolic Regression for Data-Driven Equation Discovery (2024)	Singireddy	Demonstrated symbolic regression for discovering mathematical relationships from data.	Provided implementation insights for equation discovery using SR.

8	Symbolic Regression Methods for Reinforcement Learning (2021)	Kubalík et al.	Used symbolic regression to learn interpretable value functions in reinforcement learning systems.	Demonstrates SR applicability across complex learning tasks.
9	Fuzzy Emulated Symbolic Regression (2022)	Beyhan	Integrated fuzzy logic with symbolic regression to handle uncertainty in data-driven modeling.	Inspired handling nonlinear relationships and uncertainties in models.
10	Distilling Free-Form Natural Laws from Experimental Data (2009)	Schmidt & Lipson	Early work on discovering physical laws using symbolic regression algorithms.	Demonstrates SR capability for discovering interpretable physical relationships.
11	Machine Learning for Estimation of Building Energy Consumption (2018)	Seyedzadeh et al.	Applied machine learning techniques for predicting building energy demand.	Supports the ML modeling component of our framework.

12	Applications of Artificial Intelligence for Energy Efficiency in Buildings (2020)	Ahmad et al.	Review of AI techniques used in building energy efficiency prediction and optimization.	Provides background for AI-based energy modeling.
13	Extrapolation and Learning Equations (2017)	Martius & Lampert	Proposed neural architectures capable of learning mathematical equations from data.	Related to interpretable equation learning approaches.
14	Controlling Bloat in Symbolic Regression Using Pareto Optimization (2019)	La Cava et al.	Introduced methods to control equation complexity while maintaining accuracy.	Inspired complexity vs accuracy analysis in our work.
15	Learning Feature Engineering for Classification Using Genetic Programming (2016)	La Cava et al.	Used genetic programming to automatically discover useful feature transformations.	Demonstrates evolutionary search techniques used in SR frameworks.

Problem Statement

Modern buildings account for a significant portion of **global energy consumption**, making accurate **heating load** prediction essential for **energy-efficient** design and sustainable infrastructure planning. Although traditional machine learning models such as **linear regression** and **ensemble methods** can achieve strong predictive performance, they often operate as **black-box** systems. This lack of interpretability limits their usefulness in engineering decision-making, where understanding the influence of structural parameters on energy demand is crucial.

There is a need for a **modeling framework** that not only predicts heating load accurately but also provides explicit **mathematical relationships** between **building features** and **energy consumption**. The problem addressed in this project is the development of an **interpretable** and deployable **symbolic regression-based** system that can discover **transparent equations**, evaluate their performance against traditional models, and integrate them into a real-time simulation and optimization platform for practical engineering applications.

Objectives

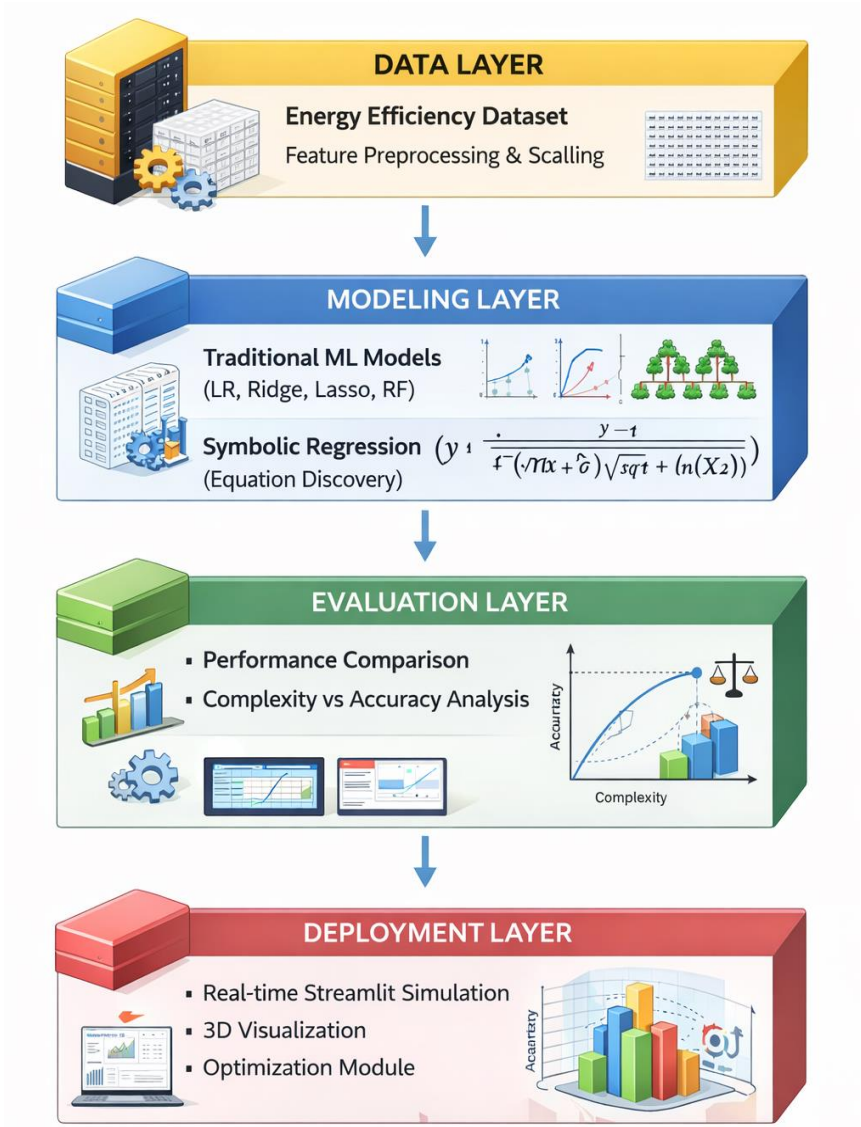
1) **To develop a predictive modeling framework** for building heating load using traditional regression techniques and evaluate their performance.
2) **To apply symbolic regression for interpretable equation discovery**, enabling the extraction of explicit mathematical relationships between building design parameters and heating load.
3) **To analyze the trade-off between model accuracy and interpretability** by comparing symbolic regression models with conventional machine learning approaches.
4) **To design and deploy a real-time interactive simulation and optimization platform** that visualizes model behavior, performs sensitivity analysis, and identifies optimal building parameter configurations for energy efficiency.

Proposed Methodology

I. SYSTEM ARCHITECTURE

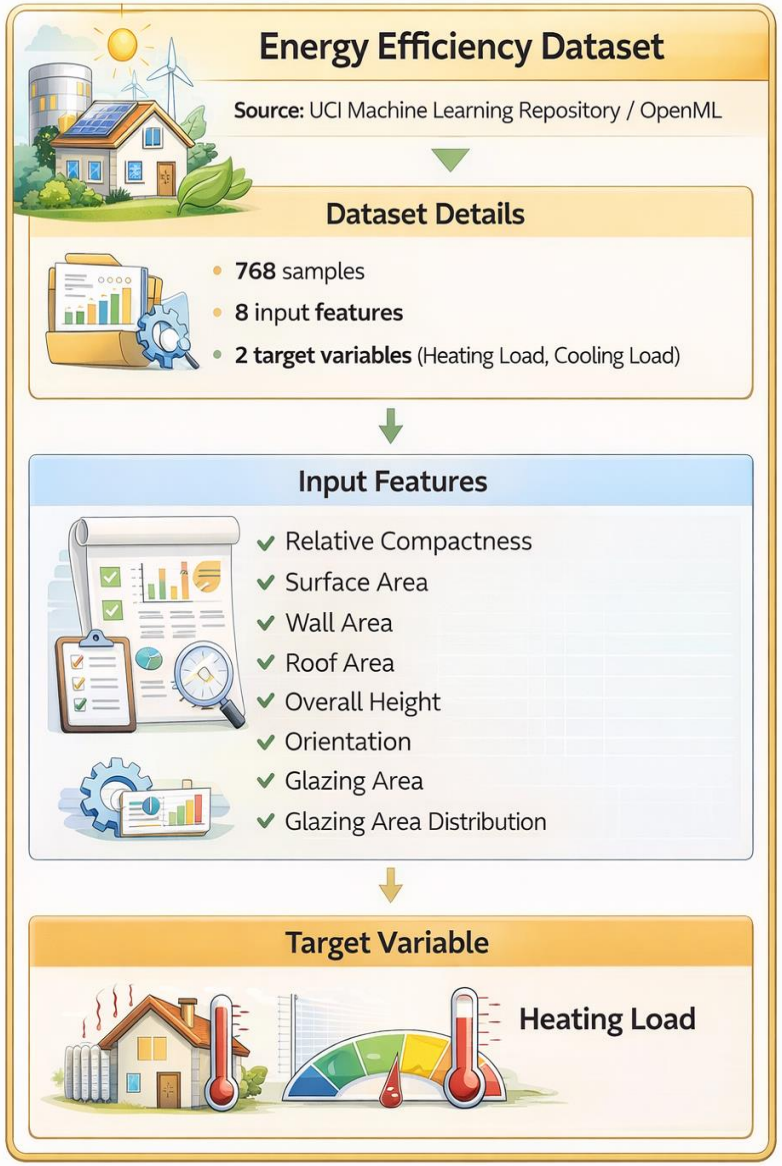
The system consists of the following major components:

- **Data Layer**
  - Energy Efficiency dataset
  - Feature preprocessing and scaling
- **Modeling Layer**
  - Traditional ML models (LR, Ridge, Lasso, RF)
  - Symbolic Regression (Equation Discovery)
- **Evaluation Layer**
  - Performance comparison
  - Complexity vs accuracy analysis
- **Deployment Layer**
  - Real-time Streamlit simulation
  - 3D visualization
  - Optimization module

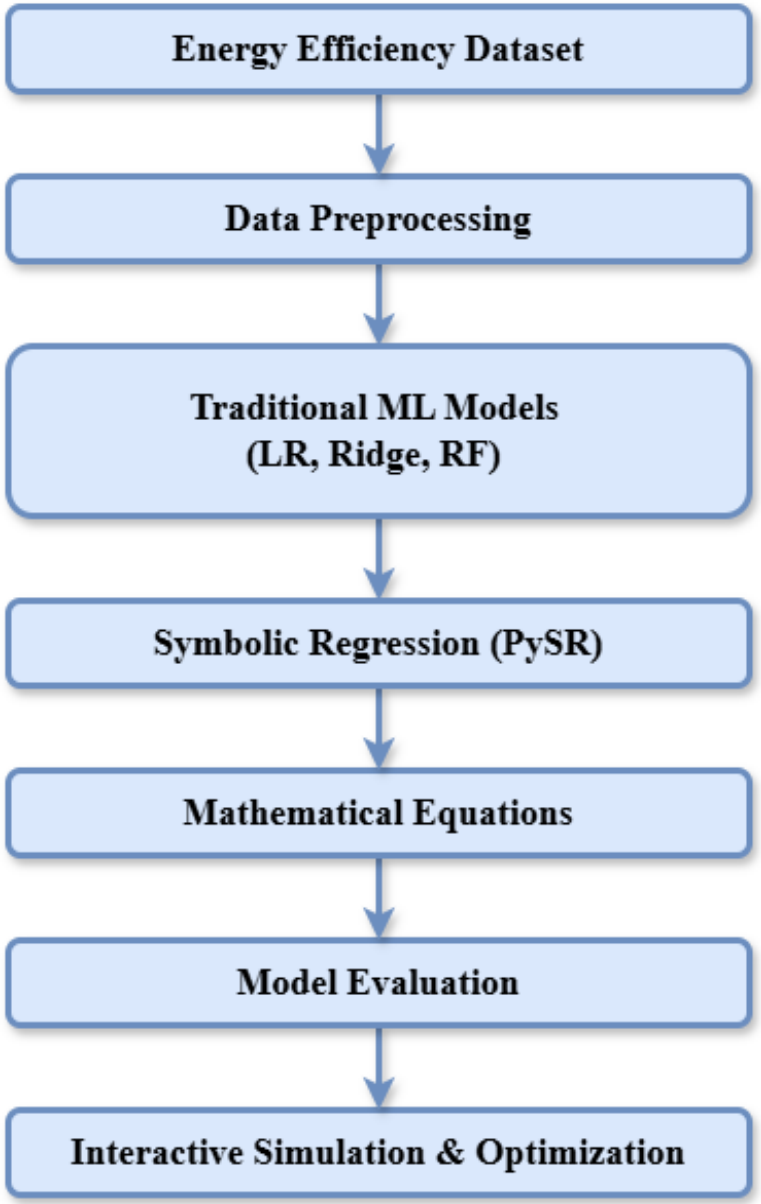


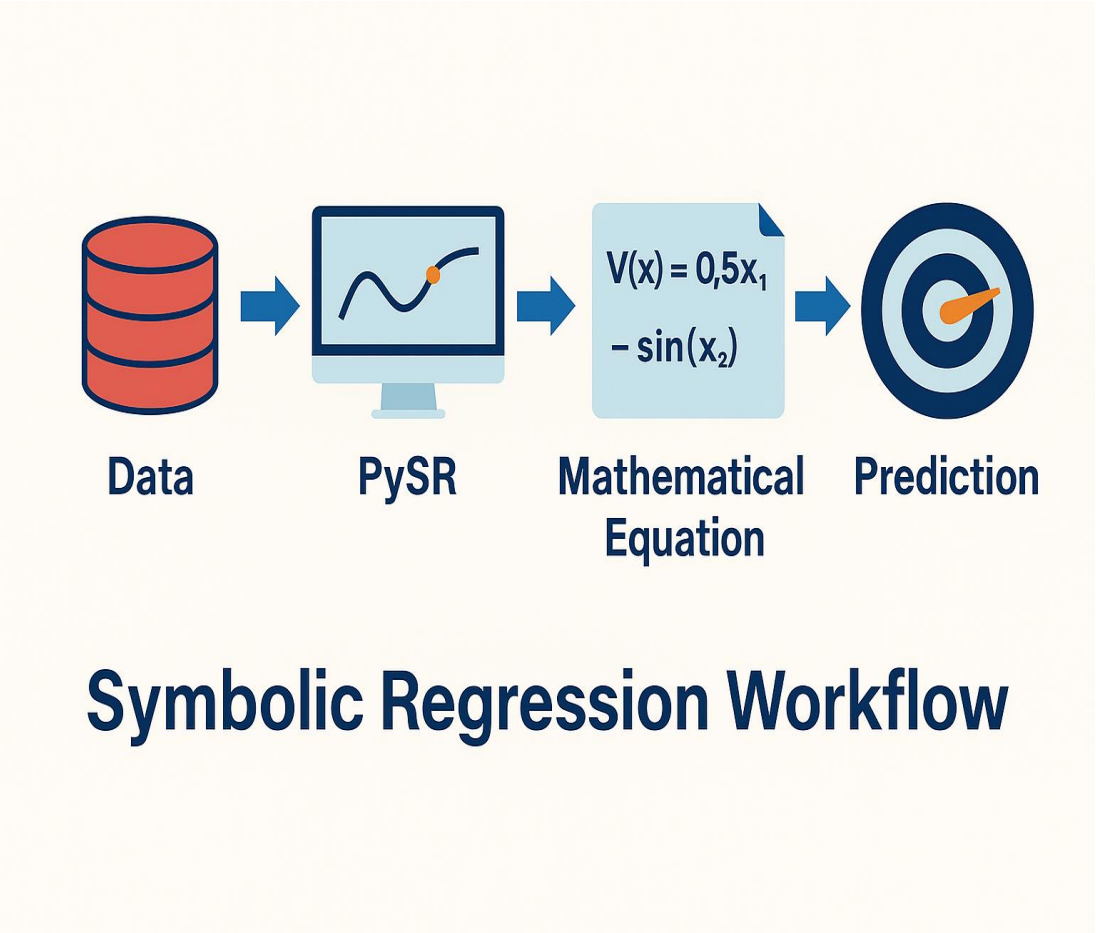
II. DATASET USED

- **Dataset Name:** Energy Efficiency Dataset
  Source: UCI Machine Learning Repository / OpenML
- **Dataset Details:**
  - 768 samples
  - 8 input features
  - 2 target variables (Heating Load, Cooling Load)
- **Input Features:**
  - Relative Compactness
  - Surface Area
  - Wall Area
  - Roof Area
  - Overall Height
  - Orientation
  - Glazing Area
  - Glazing Area Distribution
- **Target Variable:**
  - Heating Load



III. System Architecture Workflow





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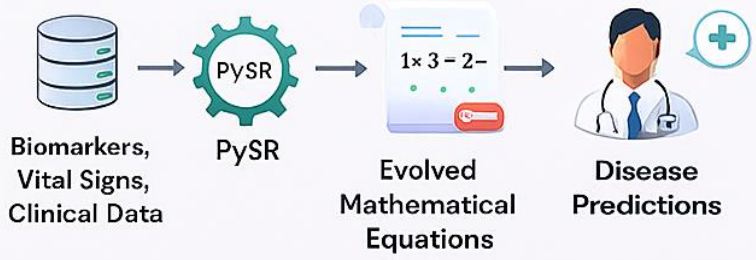
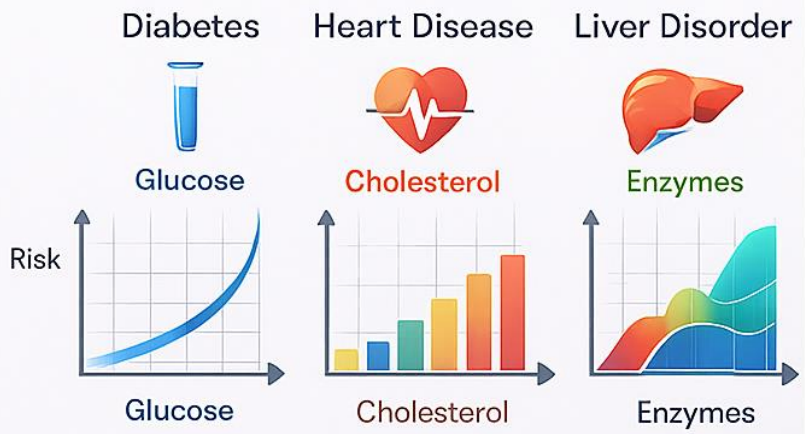
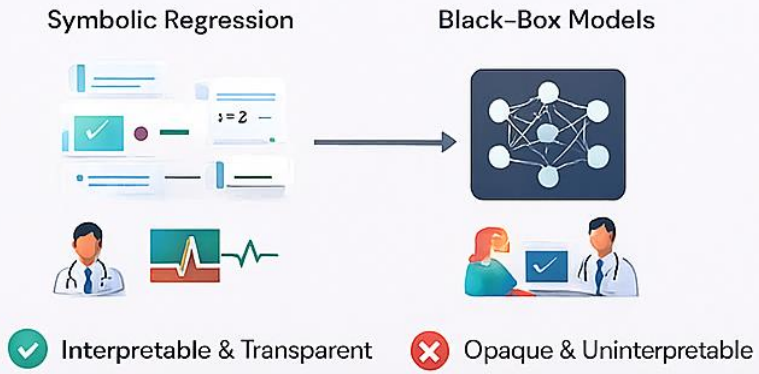
Symbolic Regression for Disease Prediction

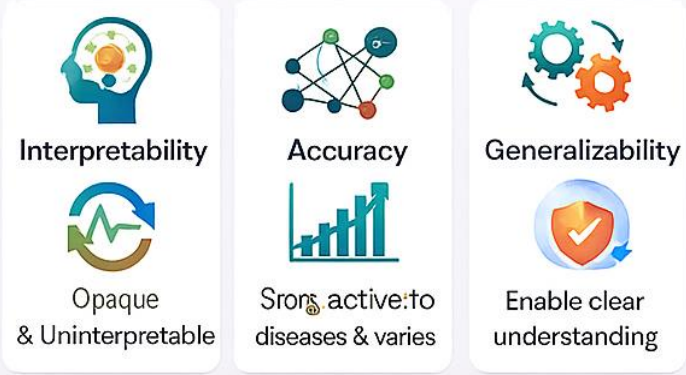




SR vs Black-Box Models in Healthcare



Benefits of Symbolic Regression



TRADITIONAL ML MODELS:

To establish baseline predictive performance, multiple regression models were implemented on the Energy Efficiency dataset.

- **Models Used:**
  - Linear Regression
  - Ridge Regression
  - Lasso Regression
  - Random Forest Regressor
- **Evaluation Metrics:**
  - R^2 Score
  - Root Mean Squared Error (RMSE)
  - Mean Absolute Error (MAE)

These models were trained on preprocessed and scaled features to evaluate their ability to predict heating load accurately.

Results Table:

Model	R^2 Score	RMSE	MAE
Linear Regression	0.9118038162167849	3.0319796709720896	2.17221658932295
Ridge Regression	0.9116028949393403	3.0354313093878527	2.1907793299161833
Lasso Regression	0.9112858626977701	3.040869656590189	2.204965713990391
Random Forest Regression	0.9934775756360946	0.8245289599573794	0.4713533817734691

SYMBOLIC REGRESSION MODEL:

Symbolic Regression (SR) was applied to discover an explicit mathematical relationship between building design parameters and heating load. Unlike traditional models, SR searches the space of mathematical expressions using evolutionary optimization and produces human-readable equations.

- **Best Discovered Equation:**

$$\textbf{Heating Load} = x2 + (x4 + 2.5169373) * (x6 + 8.7142315 + 0.24151891 / (0.38839805 - x0))$$

Where:

- x_0 =Relative Compactness
- x_2 =Surface Area
- x_4 =Roof Area
- x_6 =Glazing Area

- **Key Observations**

  - Symbolic Regression achieves **high performance** with **interpretable equations**.
  - The model captures **nonlinear relationships** between features.
  - **Surface area** and **roof area** have major influence on **heating load**.
  - Provides a balance between **accuracy** and **explainability**.

Equation Discovery Process:

Symbolic Regression generates multiple candidate equations of increasing complexity and evaluates them using a loss function.

- The search process optimizes:
  1. Prediction error (Loss)
  2. Fitness score
  3. Equation complexity
  4. Mathematical simplicity

- Evolution of Best Equations:

Complexity	Loss	Equation (Simplified)
1	100.92	(y = 22.15)
3	84.05	(y = x4 + 22.15)
4	21.22	(y = (x4 + 4.59)^2)
6	16.73	(y = x6 + (x4 + 4.59)^2)
10	8.50	Nonlinear interaction of (x4, x6)
12	6.07	Interaction with (x2, x4, x6)
15 (Best)	2.60	Final optimized equation

Results Table:

Model	R^2 Score	RMSE	MAE
Symbolic Regression	0.9707777473317559	1.7452523600481433	1.31030150996022

Why Symbolic Regression Outperforms Traditional ML

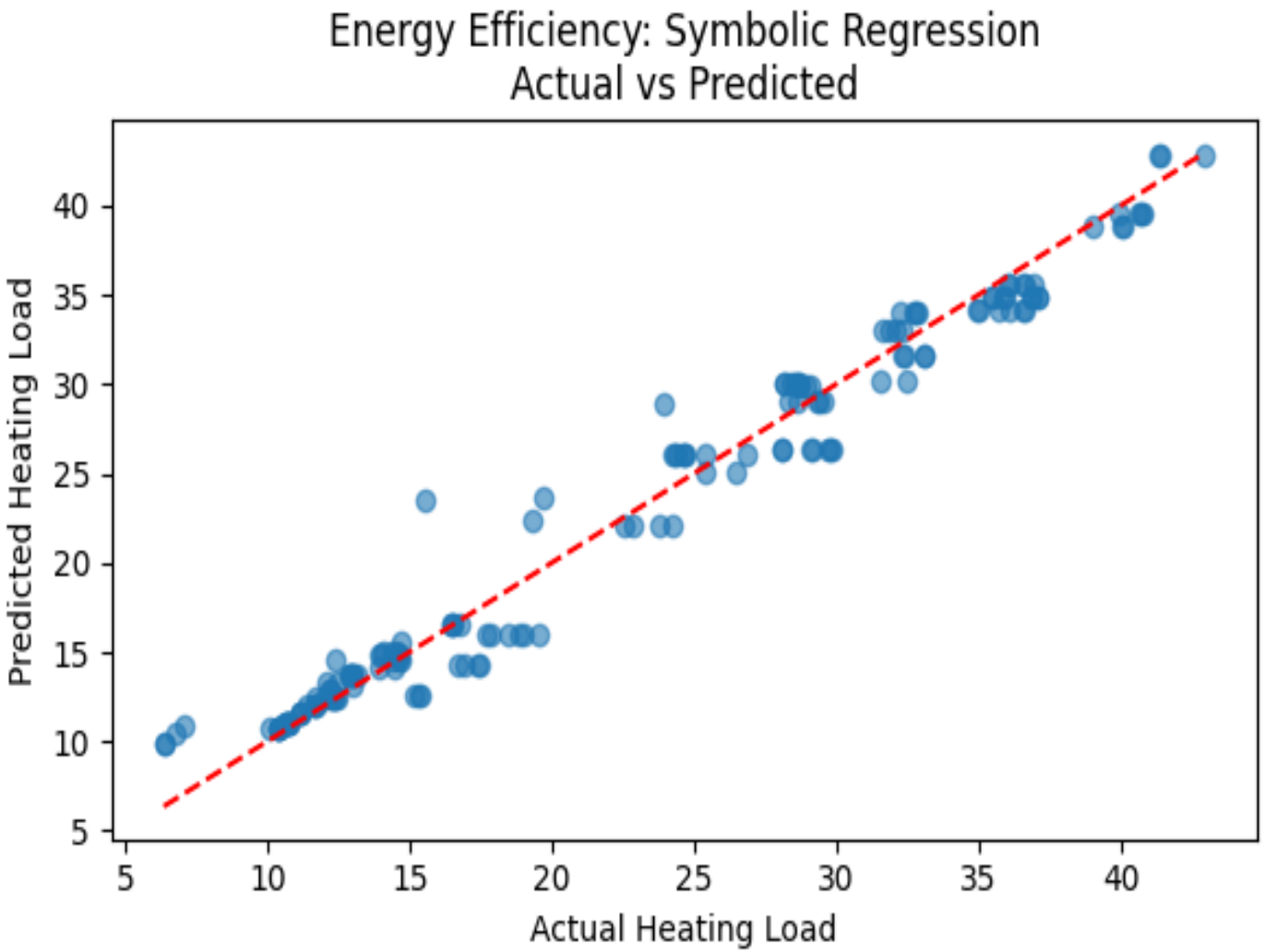
- Provides an explicit mathematical equation instead of a black-box prediction.
- Reveals clear relationships between building parameters and heating load.
- Enables direct analytical interpretation and engineering insight.
- Supports optimization using the derived equation.
- Maintains competitive predictive accuracy compared to traditional ML.
- Lightweight and easily deployable in real-time simulation systems.

Comparison Table:

Aspect	Traditional ML (LR, RF, etc.)	Symbolic Regression (SR)
Model Type	**Statistical / Ensemble-based**	**Evolutionary equation discovery**
Interpretability	Low (especially RF, ensembles)	High (explicit mathematical formula)
Output	Numerical prediction	Mathematical equation + prediction
Transparency	Limited (black-box behavior)	Fully transparent
Feature Relationship	Implicit	Explicit mathematical relationship
Accuracy	Often high	Competitive, slightly lower in some cases
Deployment	Easy	Easy after equation extraction
Computational Cost	Moderate	Higher during training
Use in Engineering	Limited explainability	Strong interpretability
Insight Generation	Minimal	Reveals nonlinear variable influence

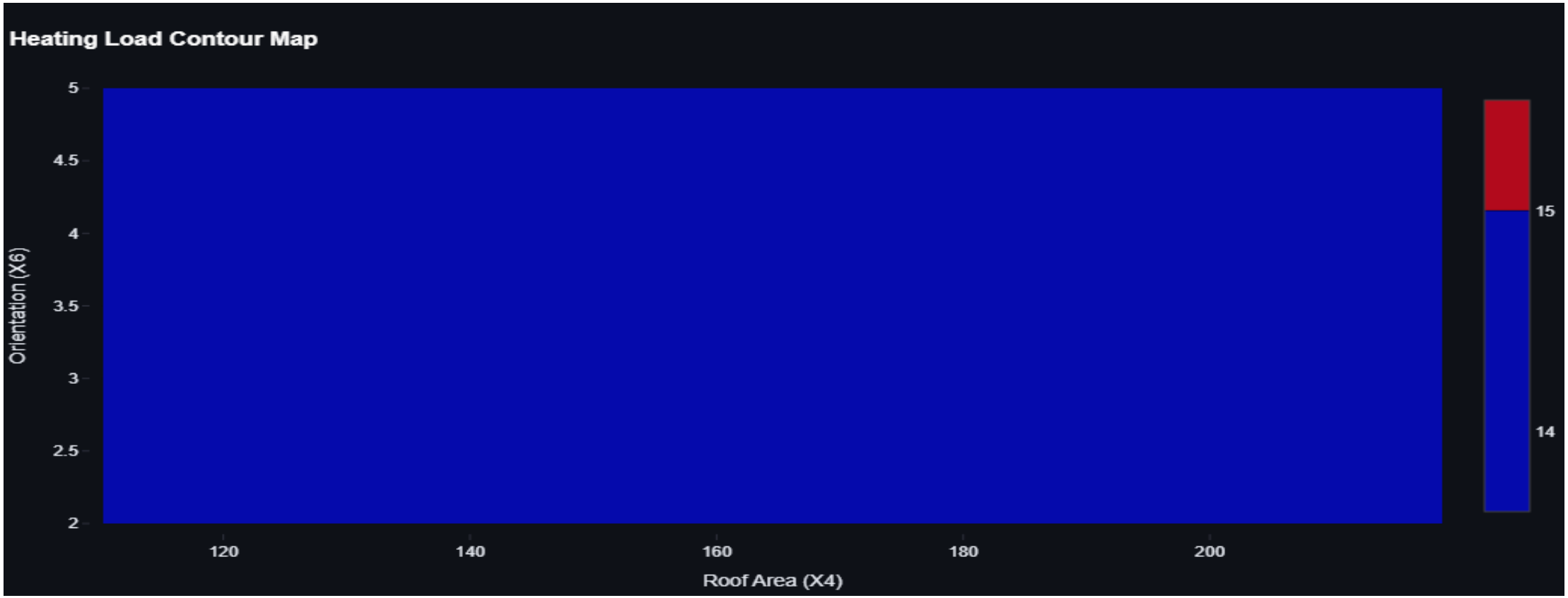
Model Validation: Actual vs Predicted (SR):

- To evaluate how accurately the symbolic regression model predicts heating load compared to actual values.
- **Description of the Plot**
  - X-axis → Actual Heating Load
  - Y-axis → Predicted Heating Load
  - Blue dots → Model predictions
  - Red dashed line → Perfect prediction reference line



Contour Map of Heating Load:

- Top-down representation of heating load distribution
- Nearly uniform color indicates stable heating load across inputs
- Minimal variation shows weak influence of roof area and orientation
- Helps identify stable design regions with consistent energy performance



Feature Importance Analysis:

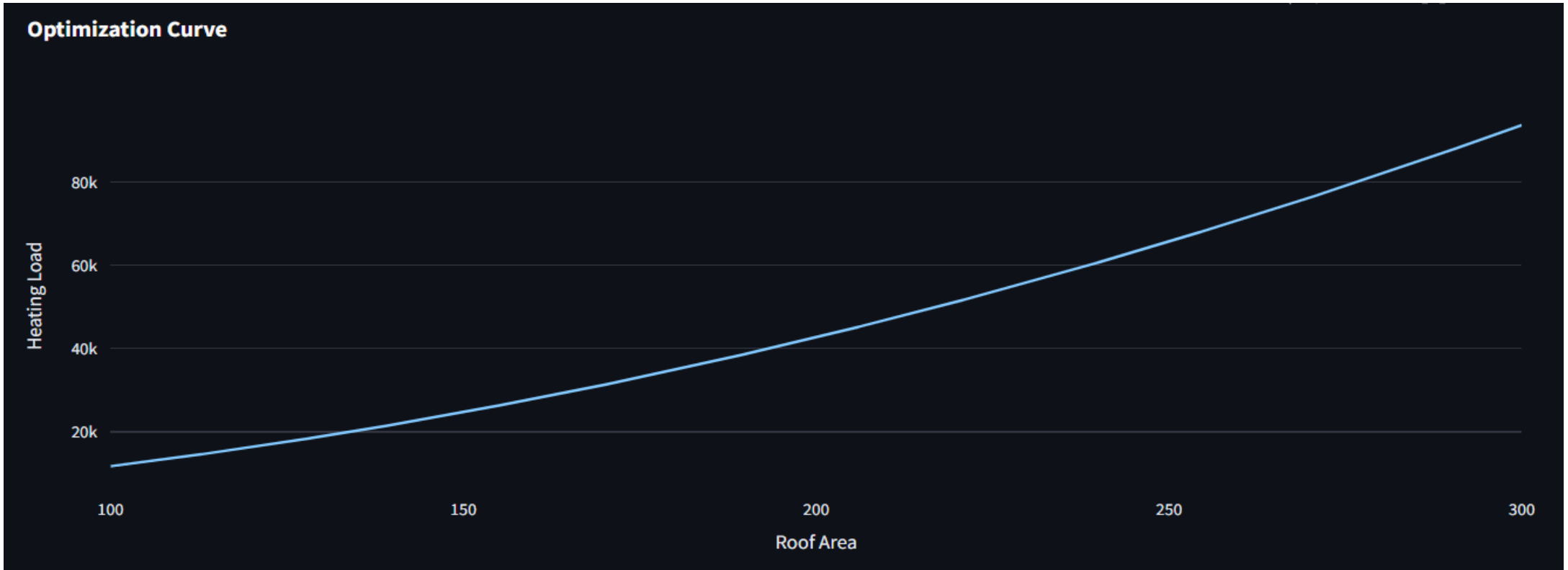
- Relative Compactness (x_1) → Most significant feature (~0.40)
- Surface Area (x_2) → Second highest impact (~0.20)
- Roof Area (x_4) and Overall-Height (x_5) → Moderate influence (~0.13)
- Orientation (x_6) → Negligible impact (≈ 0.00), confirming weak contribution

Heating Load Optimization:

To minimize heating load by identifying the optimal building parameter configuration using the discovered symbolic regression equation.

- **Optimization Variable**
  - **Roof Area (x_4)**
  - Other parameters kept constant
  - Symbolic equation used for analytical optimization

- **Optimization Result**
  - Optimal Roof Area: **100.00**
  - Minimum Heating Load: **11724.16**



Conclusion

The proposed **Symbolic Regression–based framework** effectively demonstrates how **interpretable and mathematically transparent models** can be applied to biomedical disease prediction. By using SR on datasets related to **Diabetes, Heart Disease, and Liver Disorders**, the system identifies explicit equations that show how clinical variables such as **glucose**, **cholesterol**, **BMI**, and **enzyme levels** influence disease outcomes. This ensures **high interpretability**, making the results understandable and meaningful for clinicians, unlike traditional **black-box machine learning models**.

Overall, the project highlights the potential of Symbolic Regression as a **trustworthy, generalizable, and clinically relevant** method for healthcare analytics. The framework not only achieves **competitive predictive performance**, but also provides **human-readable insights** that support better decision-making. This work shows that SR can bridge the gap between **accuracy** and **explainability**, enabling the development of reliable AI systems suitable for real-world medical applications.

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